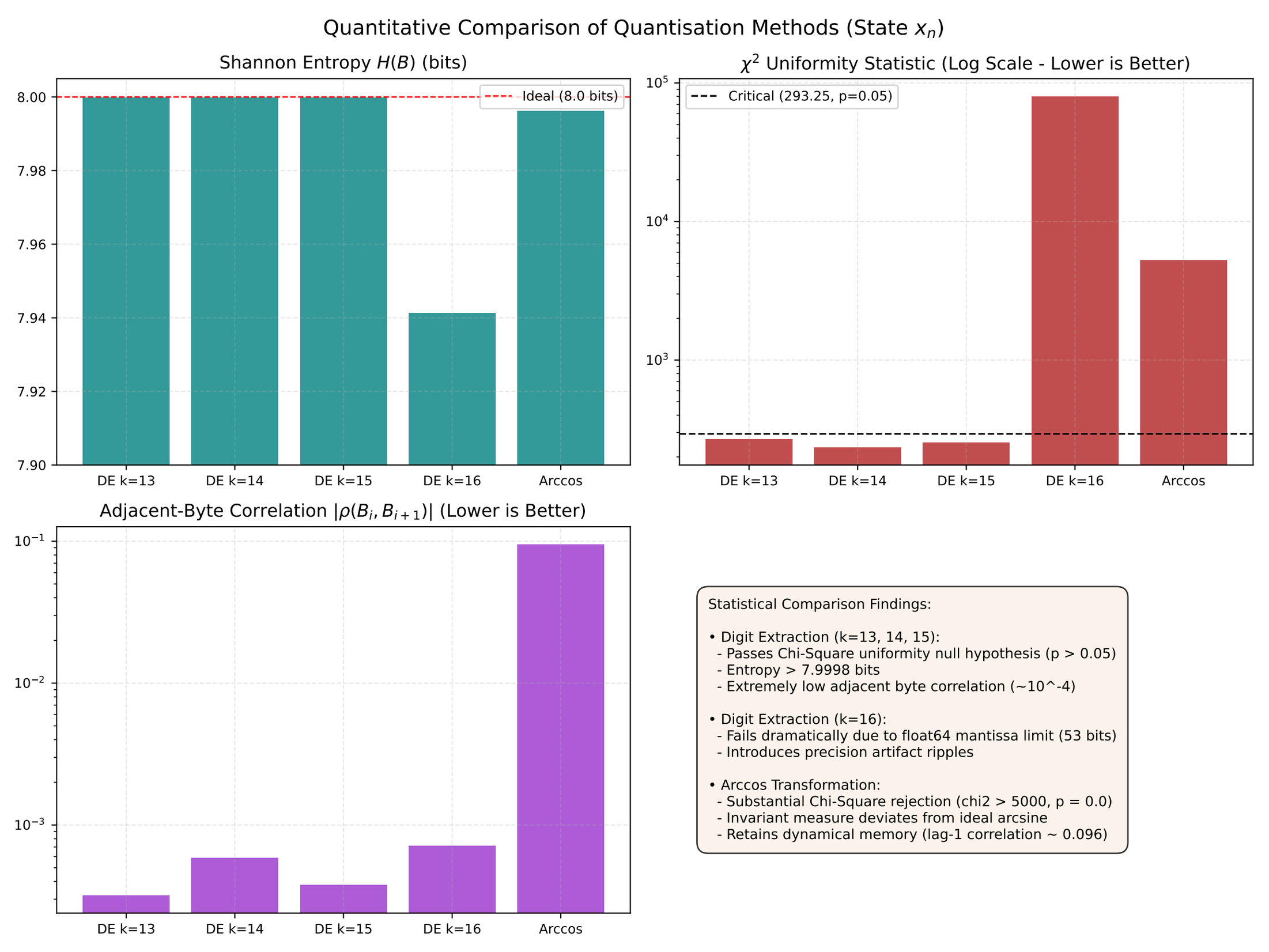
Quantitative Comparison of Quantisation Methods (State x_n)
Shannon Entropy $H(B)$ (bits)
Ideal (8.0 bits)
8.00
7.98
7.96
7.94
7.92
7.90
DE k=13
DE k=14
DE k=15
DE k=16
Arccos
χ^2 Uniformity Statistic (Log Scale - Lower is Better)
Critical (293.25, p=0.05)
10^5
10^4
10^3
DE k=13
DE k=14
DE k=15
DE k=16
Arccos
Adjacent-Byte Correlation $|\rho(B_i, B_{i+1})|$ (Lower is Better)
10^{-1}
10^{-2}
10^{-3}
DE k=13
DE k=14
DE k=15
DE k=16
Arccos
Statistical Comparison Findings:
• Digit Extraction (k=13, 14, 15):
- Passes Chi-Square uniformity null hypothesis (p > 0.05)
- Entropy > 7.9998 bits
- Extremely low adjacent byte correlation (~10^-4)
• Digit Extraction (k=16):
- Fails dramatically due to float64 mantissa limit (53 bits)
- Introduces precision artifact ripples
• Arccos Transformation:
- Substantial Chi-Square rejection (chi2 > 5000, p = 0.0)
- Invariant measure deviates from ideal arcsine
- Retains dynamical memory (lag-1 correlation ~ 0.096)